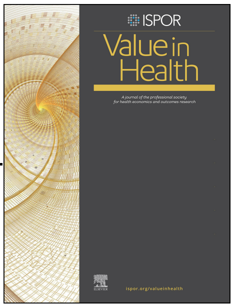
Journal Pre-proof



A comparative study of health and wellbeing measures in Ireland using EQ-HWB, EQ-HWB-S, EQ-5D-5L, and ICECAP-A

Irina Kinchin, PhD, Lidia Engel, PhD, Fanni Rencz, PhD

PII: S1098-3015(25)02324-1

DOI: https://doi.org/10.1016/j.jval.2025.04.2160

Reference: JVAL 4320

To appear in: *Value in Health*

Received Date: 9 October 2024

Revised Date: 10 March 2025

Accepted Date: 12 April 2025

A comparative study of health and wellbeing measures in Ireland using EQ-HWB, EQ-HWB-S, EQ-5D-5L, and ICECAP-A

Irina Kinchin PhD[1,*], Lidia Engel PhD[2], Fanni Rencz PhD[3]

[1] Centre for Health Policy and Management, School of Medicine, Trinity College Dublin, Dublin, Ireland

[2] Monash University Health Economics Group, School of Public Health and Preventive Medicine, Monash University, Clayton, Victoria, Australia

[3] Department of Health Policy, Corvinus University of Budapest, Budapest, Hungary

***Corresponding author:** Dr Irina Kinchin, Centre for Health Policy and Management, Trinity College Dublin, Dublin 2, Ireland Email: irina.kinchin@tcd.ie

Funding: Irina Kinchin is supported by the Health Research Board (HRB) Fellowship 'Applying Research into Policy and Practice (ARPP)' (ARPP-2020-011) and the Applied Programme Grants (APRO) Award (Reference 9745). The EuroQol Research Foundation supported secondary analysis of the data collected as part of the HRB ARPP (ARPP-2020-011) (EQ Project 1676-RA). The funding sources had no involvement in the study design; in the collection, analysis, and interpretation of data; in the writing; or in the decision to submit the article for publication.

Précis: EQ-HWB shows solid psychometric properties in Ireland, also in comparison with EQ-5D-5L and ICECAP-A, which will be valuable for informing decision-making about healthcare and social care interventions.

Authorship contribution: All authors certify that they meet the ICMJE criteria for authorship.

Word count: 4221

Number of pages: 35

Number of figures: 0

Number of tables: 6

Supplementary files: 1

Dr Lidia Engel is an editor for Value in Health and had no role in the peer-review process of this article.

A comparative study of health and wellbeing measures in Ireland using EQ-HWB, EQ-HWB-S, EQ-5D-5L, and ICECAP-A

ABSTRACT

Objectives: The EQ Health and Wellbeing (EQ-HWB) is a new experimental measure of health and wellbeing, having been validated in an increasing number of countries and languages. This study aims to examine the psychometric properties of the EQ-HWB and its short version (EQ-HWB-S) in Ireland and compare them to the EQ-5D-5L and ICECAP-A.

Methods: A secondary analysis of cross-sectional data from a 2023 Irish general population survey (*n*=1220). The survey included the EQ-HWB, EQ-5D-5L, ICECAP-A, and socioeconomic and health-related questions. EQ-HWB-S responses were derived from the EQ-HWB. The UK or Irish value sets were used for each measure to compute index values. Ceiling effects, convergent, divergent validity, known-group validity were assessed, exploratory factor analysis was conducted.

Results: Mean index values were 0.755, 0.760-0.830, and 0.807 for the EQ-HWB-S, EQ-5D-5L, and ICECAP-A, respectively. Ceiling effects were 3%, 7%, 23%, and 15% for EQ-HWB, EQ-HWB-S, EQ-5D-5L, and ICECAP-A. EQ-HWB-S index values correlated strongly with EQ-5D-5L (r=0.740-0.759) and ICECAP-A (r=0.604) index values. All measures demonstrated good known-group validity for health-related variables, with EQ-5D-5L performing best overall. Factor analysis identified five factors: psychosocial health, pain and discomfort, sensory and physical functioning, capability wellbeing, and positive psychological states. Stability was the only ICECAP-A item that loaded on the same factor (psychosocial health) as any EQ-HWB items.

Conclusion: This is the first study to compare the measurement performance of EQ-HWB, EQ-HWB-S with ICECAP-A. The EQ-HWB and EQ-HWB-S showed good psychometric performance in an Irish population sample. Limited overlap was observed between EQ-HWB and ICECAP-A wellbeing concepts.

Keywords: EQ-HWB, EQ-HWB-S, ICECAP-A, EQ-5D-5L, health-related quality of life, capability wellbeing, psychometric analysis

HIGHLIGHTS

i. What is already known about the topic? EQ-HWB and its shorter version (EQ-HWB-S) are promising measures for evaluating interventions in healthcare, public health, and social care. More evidence is needed to understand how they perform and relate to established measures such as EQ-5D-5L and ICECAP-A.

ii. What does the paper add to existing knowledge? Both the EQ-HWB and EQ-HWB-S demonstrate robust psychometrics in an Irish general population sample, with lower ceiling effects than EQ-5D-5L and ICECAP-A, showing a distinct factor structure for health and capability wellbeing concepts. The EQ-HWB and EQ-HWB-S perform better or similarly to ICECAP-A in differentiating between known groups of respondents.

iii. What insights does the paper provide for informing healthcare-related decision making? EQ-HWB and EQ-HWB-S capture broader wellbeing aspects than EQ-5D-5L while supplementing ICECAP-A, offering decision-makers a nuanced tool for health and wellbeing assessment in clinical and policy landscapes.

INTRODUCTION

The EQ Health and Wellbeing (EQ-HWB) instrument has been developed to assess outcomes of interventions in healthcare, public health, and social care for use among patients, service users and caregivers.[1] It aims to measure quality of life but extends the evaluative space beyond health-related quality of life, capturing aspects of wellbeing while still allowing for the estimation of quality-adjusted life-years (QALYs) to inform decision-making both within and between these sectors.[2] The EQ-HWB has a long-form (25 items) and a short form representing a subset of items (9 items; EQ-HWB-S), each serving different purposes. The long form is a profile measure, while the short form functions as a health classifier with a preference-based scoring (i.e. value set).[3] The EQ-HWB-S has been officially recognized in Dutch health technology assessment guidelines as a recommended tool for evaluating the benefits of non-curative interventions.[4]

The capability approach and associated ICECAP measures [ICECAP-A (ICEpop CAPability measure for Adults 18+), ICECAP-O (ICEpop CAPability measure for Older people 65+), ICECAP-SCM (ICEpop CAPability Supportive Care Measure for use in economic evaluation conducted in an end of life setting)] provide an alternative framework for conceptualizing and measuring wellbeing, which differs from the EQ-HWB in several ways.[5-7] The capability approach emphasizes the importance of individual agency and freedom to pursue life goals, and the ICECAP measures focus on capability and broader domains of wellbeing, such as autonomy, social participation, and security.[8] The ICECAP-A and ICECAP-O demonstrated good measurement performance in multiple populations.[9, 10] In some countries, including the UK[11] and the Netherlands,[4] the ICECAP measures are on the list of health technology assessment agencies of recommended instruments to measure outcomes of health interventions with potential effects beyond health, e.g., social or long-term care.

While both the EQ-HWB-S and ICECAP measures employ preference-based scoring, there are important methodological distinctions in their approach. EQ-HWB-S values are anchored on the traditional utility scale, with 0 representing 'dead' and 1 representing 'full health'. Utilities can be multiplied with life years (i.e. survival time) to compute QALYs for cost-utility analysis. Conversely, ICECAP values are anchored on a 'no capability' (0) to 'full capability' (1) scale. In cost-effectiveness analyses, ICECAP values can be combined with time to generate years of full capability (YFCs), representing the total available capability over time.[12, 13]

The current evidence on how well the EQ-HWB performs compared to other wellbeing or social care-related measures is limited. Existing qualitative and quantitative validation studies drew comparisons to the short Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS), personal-wellbeing (PWI-A), CarerQol-7D, Quality of Life-Aged Care Consumers (QOL-ACC) and Adult Social Care Outcomes Toolkit (ASCOT).[14-16] However, no studies have compared the measurement performance of the EQ-HWB with any ICECAP measures to date.

Building on recent advances in measurement theory that emphasize the contextual nature of validity evidence[17], and recognizing that measurement support varies across different settings and populations, this study addresses important knowledge gaps by: (1) providing the first comparative analysis of the EQ-HWB and ICECAP-A, offering insights for instrument selection in health and social care evaluation; and, (2) establishing the validity of these measures in Ireland's distinct healthcare context, where system structures and societal values may differ from previously studied populations.

METHODS

Study design and sample

This study involved a secondary analysis of data collected from a cross-sectional online survey that aimed to derive a value set for the ICECAP Supportive Care Measure (ICECAP-SCM) based on preferences from the general Irish population using best-worst scaling (BWS) and discrete choice experiment (DCE) tasks.[18] The 25-item long format EQ-HWB along with EQ-5D-5L and ICECAP-A were included in the survey. The survey opened on 20 February 2023 and closed on 05 April 2023.

SurveyEngine, an international survey company specializing in choice experiments and complex survey designs, recruited a sample of the adult general population in Ireland with nationally representative quotas on gender and age using their panel members. Panel members are pre-recruited individuals who complete demographic profiles during registration and participate regularly in research studies. They undergo identity verification and quality checks.

Survey

The survey was in English, self-administered online and included sections on socio-demographic characteristics (age group, gender, partnership status, caregiving status, dependent children, general health status on excellent-to-poor scale with reasons in open text box, ongoing medical conditions, country of birth and current county, the importance of religion and spiritual beliefs, ethnicity, education and work status, experience of a life-limiting illness); 16 DCE and 16 BWS choice tasks based on ICECAP-SCM; the EQ-5D-5L, the ICECAP-A and the 25-item EQ-HWB 25-item. Ongoing medical conditions were assessed with the following question: *Do you have any of the following medical conditions? Select all that apply*. Caregiving status was assessed with the following two questions: *Are you currently providing help or support to a family member, friend, or neighbour (adult or child) who has a disability, mental health difficulty, chronic condition, dementia, terminal illness, drug or alcohol dependency, who needs care due to aging? If yes, how many hours did you spend on* *helping or supporting this person in the last 7 days*. The order of EQ-5D-5L, the ICECAP-A and the EQ-HWB was fixed. All questions in the survey were mandatory; and for some questions there was a ‘prefer not to answer’ option, therefore there were no missing data.

Outcome measures

The survey collected data on participant health and wellbeing using experimental English version for the UK of the EQ-HWB (version 1.2), English digital version of the EQ-5D-5L for the UK (version 1.0) and English version of the ICECAP-A.

EQ-HWB: The 25-item EQ-HWB captures health and well-being across domains of activity, autonomy, cognition, feelings and emotions, relationships, physical sensations, and self-identity [2]. Except for the self-identity domain, the EQ-HWB-S captures the same domains as the full EQ-HWB. The timeframe is the past seven days. In our study, participant responses for the 9-item EQ-HWB-S were extracted from the EQ-HWB.

Responses to items in the EQ-HWB are captured on a 5-point frequency scale (“None of the time,” “Only occasionally,” “Sometimes,” “Often,” “Most or all of the time”), difficulty scale (“No difficulty,” “Slight,” “Some,” “A lot of,” “Unable”), or severity scale (“no physical pain,” “mild physical pain,” “moderate physical pain,” “severe physical pain,” “very severe physical pain”). For the EQ-HWB, level sum scores (LSS) were computed. Prior to this computation, responses to the three positively framed items ("Felt accepted," "Felt good about myself," and "I could do the things I wanted to do") were reverse scored to ensure consistency with the other items, where higher scores indicate worse health or wellbeing. This reverse scoring was necessary to maintain a consistent interpretation across all items, where higher scores consistently represent poorer outcomes. After computing the LSS, scores were transformed to a 0-100 range to facilitate interpretation and comparison. In this transformed scale, lower scores indicate better health and wellbeing. We also used recently developed UK pilot value set for the EQ-HWB-S to obtain index values [3]. Potential values range from -0.384 (worst level, problems on all items) to 1 (full health and quality of life). The EQ-HWB was presented in blocks of items, maintaining consistency with the paper-based version format.

EQ-5D-5L: The EQ-5D-5L is a generic measure of health that comprises of two parts: a descriptive system and the EQ VAS.[19] The descriptive system captures five dimensions of health, these are mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension uses a severity scale with participants' responses ranging from 'no problems' (1) to 'unable to'/'extreme problems' (5). The EQ VAS records self-rated health on a vertical health thermometer anchored at 0 ('the worst health you can imagine') to 100 ('the best health you can imagine'). The timeframe for both the descriptive system and EQ VAS is the day of the completion ('today'). Multiple EQ-5D-5L value sets were used for the comparisons conducted in this study, including the Irish EQ-5D-5L index value set (values ranging from -0.974 (extreme problems on all dimensions) to 1 (full health)),[20] the English value set (values ranging from -0.285 to 1),[21] and the UK values derived using crosswalk from the EQ-5D-3L (values ranging from -0.594 to 1).[22]

ICECAP-A: The ICECAP-A is a measure of capability wellbeing for adults (18+) consisting of the following five single-item dimensions: stability (an ability to feel settled and secure), attachment (an ability to have love, friendship and support), autonomy (an ability to be independent), achievement (an ability to achieve and progress in life) and enjoyment (an ability to experience enjoyment and pleasure).[7] In each dimension, participants' responses may range from 'no capability' (1) to 'full capability' (4). The timeframe is 'at the moment'. The UK value set was used to generate ICECAP-A scores.[23] Potential values range from 0 to 1, where 0 is the worst (i.e. no capability on any dimension) and 1 the best (full capability on all dimensions).

Statistical analyses

Participant characteristics

Descriptive statistics were used to describe key characteristics of the surveyed sample. Continuous variables were expressed as mean and SD or median and mode where relevant, whereas categorical variables were expressed as frequencies and percentages.

Floor and ceiling

We examined both the item-level and instrument-level floor and ceiling effects of the EQ-HWB and EQ-HWB-S, and drew comparisons to the EQ-5D-5L and ICECAP-A. We considered a ceiling or floor effect at the item level to be present if more than 70% of respondents answered in the top or bottom category for any given item.[24] However, given the general population sample, a skewed distribution was expected for several items. We used the threshold of 15% for the instrument level effect.[25] We hypothesized the highest ceiling for the EQ-5D-5L, which showed an average ceiling of 49% in earlier general population samples,[26] and the lowest for the EQ-HWB, which has the highest number of items.

Convergent and divergent validity

Spearman's and Pearson's correlations were used to assess convergent and divergent validity. Correlations were interpreted as "strong" (≥0.50), "moderate" (0.30-0.49), "weak" (0.10-0.29), and "none" (0-0.09).[27] We expected moderate or strong correlations between items of different instruments covering the same health or wellbeing area, for example, between the EQ-HWB 'unsupported' and 'loneliness' and ICECAP-A 'attachment'; the EQ-HWB 'control' and ICECAP-A 'autonomy'; EQ-HWB 'feeling unsafe' and ICECAP-A 'stability'; EQ-HWB 'do the things wanted to' and ICECAP-A 'enjoyment'; and EQ-HWB 'feeling good about oneself' and ICECAP-A 'achievement', among others. Otherwise, we expected weak or no correlation between items of different instruments (refer supplementary material for all expected correlations).

Known-group validity

Known-groups validity was examined by comparing groups defined by age, sex, employment, education, caregiving (yes/ not currently, in the past/ no), general health status on excellent-to-poor scale, EQ VAS (with a cutoff at 80),[24] and ongoing medical condition (none/any). Better health and wellbeing were anticipated with being employed, more educated, not being a caregiver, good or excellent general health status, an EQ VAS > 80.[28-31] No hypotheses were set regarding age due to the mixed evidence about the association between age and wellbeing[32], nor sex as a prior study in Ireland found no sex differences in EQ-5D.[31]

Differences in index values between groups were assessed using the nonparametric Mann–Whitney U test (2 groups) and Kruskal–Wallis H test (multiple groups). Effect sizes were calculated and interpreted as follows: eta squared based on the Kruskal–Wallis H-statistic “small” 0.01-0.059, “moderate” 0.06-0.139, “large” ≥0.14; Cohen’s r based on the *z* value for the Mann–Whitney U “small” 0.11-0.30, “moderate” 0.31-0.49, “large” ≥0.5.[33]

Dimensionality and structure

Building on previous work on dimensionality of EQ-5D-5L and bolt-ons,[34] we conducted an exploratory factor analysis (EFA) to better understand the dimensionality of the EQ-HWB and EQ-HBW-S. In addition to the 25 EQ-HWB items, we also included items from the EQ-5D-5L and ICECAP-A to see if the EQ-HWB measures similar or distinct underlying constructs related to health and wellbeing (i.e. loading on the same factor or not). The positively framed items of the EQ-HWB were appropriately recoded prior to analysis to ensure consistent interpretation of the factor loadings.

Initially, we performed Bartlett's test of sphericity and the Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy to assess the suitability of our data for factor analysis. The number of factors to retain was guided by examination of the scree plot and the application of the Kaiser criterion (eigenvalues > 1). Following this initial analysis, we conducted a Maximum Likelihood Factor Analysis, specifying a five-factor solution based on our preliminary findings. We applied a *promax* rotation to allow for potential correlations between factors, with a minimum loading criterion of 0.32 to determine item retention on factors (goodness of factor loadings: 0.33 to 0.44 (poor), 0.45 to 0.54 (fair), 0.55 to 0.62 (good), 0.63 to 0.70 (very good), and 0.71 (excellent)).[35] All statistics were two-sided, and $p < 0.05$ was considered statistically significant. Data were analysed with using Stata version 17.0 (StataCorp, 2021).

Ethics

This study has been approved by the Health Policy and Management and Centre for Global Health Research Ethics Committee [02/2022/02] at Trinity College Dublin.

RESULTS

Participant characteristics

After removing 181 entries which were speeders (below 1/3 of the median duration), IP duplicates, and 'flatliners', i.e. respondents who always chose the same answer in one of the BWS/DCE tasks, the survey company supplied a final sample of 1,220 completed surveys. Table 1 presents the demographic and health characteristics of the final study sample (*n*=1220). When compared to national representation, younger adults and females were slightly overrepresented. Regarding health status, 84% of participants reported their health as good or excellent, with 39% scoring above 80 on the EQ VAS. However, 69% reported having at least one ongoing medical condition, with anxiety (38%) and depression (19%) being the most prevalent. The sample also captured aspects of caregiving and family responsibilities, with 37% of participants providing help or support to others, and 53% having dependent children.

Floor and ceiling effects in response distribution

Across the various indices, respondents reported high levels of health and wellbeing (**Table 2**). None of the measures demonstrated a significant floor effect, with 0% of respondents scoring the minimum possible value across all domains for the EQ-5D-5L, EQ-HWB, EQ-HWB-S, and ICECAP-A (**Table 2**). There were two respondents that scored 0 on the EQ VAS. The EQ-5D-5L exhibited ceiling effect of 22.5%. Neither the EQ-HWB (2.6%) nor the EQ-HWB-S (6.6%) demonstrated any ceiling effect. The ICECAP-A demonstrated borderline ceiling effect with 14.6%. The EQ VAS did not show ceiling effect, with only 3.61% of respondents scoring 100.

Ceiling effects (>70% responses in the top category) were observed in several EQ-5D-5L (i.e., mobility, self-care, usual activities) and EQ-HWB items (i.e., hearing, mobility, personal care) (**Table 3**). However, these findings should be interpreted in the context of our sample characteristics - given the younger age profile of our respondents, high proportions of responses in the best category for physical function items (such as mobility and self-care) and sensory items (such as hearing) may reflect the actual health status of the sample rather than limitations in the measurement properties. The ICECAP-A did not demonstrate ceiling effects in any items. Floor effects were not observed in any of the items. The three positively framed EQ-HWB items showed the highest floor (10.57-14.10%).

Convergent and divergent validity

Item-level correlations between EQ-5D-5L and EQ-HWB items

Our analysis revealed strong correlations between conceptually similar domains of the EQ-5D-5L and EQ-HWB items, supporting their convergent validity. For example, the EQ-HWB item 'getting around inside and outside' showed a strong correlation with the EQ-5D-5L 'mobility' domain (r=0.579). The 'day-to-day activities' item from EQ-HWB also correlated strongly with EQ-5D-5L 'usual activities' (r=0.677). Additionally, the emotional well-being items 'feel anxious' and 'feel sad/depressed' from EQ-HWB correlated strongly with EQ-5D-5L 'anxiety/depression' (r=0.707 and r=0.648, respectively), which are detailed in supplementary material.

Item-level correlations between EQ-HWB and ICECAP-A items

We observed moderate correlations between similar items across the EQ-HWB and ICECAP-A, indicating convergent validity. Notably, the EQ-HWB items 'unsupported' and 'loneliness' showed moderate correlations with the ICECAP-A 'attachment' domain (r=0.440 and r=0.430, respectively). However, some expected relationships showed weaker correlations than anticipated. The EQ-HWB 'control' item correlated only weakly with ICECAP-A 'autonomy' (r=0.374). Similarly, 'feeling unsafe' from EQ-HWB showed a weak correlation with ICECAP-A 'stability' (r=0.246), and 'do the things wanted to' correlated weakly with ICECAP-A 'enjoyment' (r=0.196). The 'feeling good about oneself' item also demonstrated a weak correlation with ICECAP-A 'achievement'. These findings are further detailed in supplementary material.

Correlations between instruments

The ICECAP-A demonstrated a strong correlation with the EQ-HWB-S (r=0.604) and EQ-5D-5L (English value set) (r=0.525) and moderate correlation with the EQ VAS (r=0.461) (**Table 4**). The choice of EQ-5D-5L value set had a negligible impact on the results. The EQ-HWB-

LSS showed strong correlations with other outcomes [*r* ranging from -0.505 (EQ VAS) to -0.922 (EQ-HWB-S)].

Known-group validity

In line with our hypotheses, individuals who were employed, more educated, not caregivers, or in better general health reported better HRQoL or wellbeing on all measures (**Table 5**). The EQ-5D-5L indices (Ireland, UK crosswalk, and England) and EQ-HWB-S index consistently showed comparable or better performance than the ICECAP-A across most known-group comparisons. Overall, the ability of the EQ-HWB and EQ-HWB-S to differentiate between known groups was very similar. For age groups, only the EQ-HWB demonstrated a moderate effect size (ES=0.06), whereas all other measures showed small effect size. For employment status, both the EQ-5D-5L (ES=0.08) indices and EQ-HWB-S (ES=0.06) exhibited moderate effect sizes outperforming the ICECAP-A with its small effect size (ES=0.05). Although all measures were able to detect significant differences between respondents based on their level of education, with very small effect sizes (ES=0.01-0.02). Similarly, all measures differentiated between respondents based on caregiver roles, with small effect sizes (ES=0.01-0.04).

For general health status, the EQ-5D-5L showed moderate-to-large effect sizes (ES=0.13-0.14), slightly outperforming the EQ-HWB-S and ICECAP-A, which showed only moderate effect sizes (0.10 and 0.12, respectively). For EQ VAS groups, all measures exhibited large effect sizes (ES=0.43-0.47). The presence of ongoing medical conditions was well-differentiated by all measures, with the EQ-5D-5L indices showing the largest effect sizes (ES=0.54-0.59), followed by the EQ-HWB-S index (ES=0.47), and the ICECAP-A index (ES=0.36).

Dimensionality and the dimension structure

The EFA revealed a five-factor structure: (1) psychosocial health, (2) pain and discomfort, (3) sensory and physical functioning, (4) capability wellbeing and (5) positive psychological states (**Table 6**). The EQ-5D-5L items loaded on three factors: anxiety/depression on Factor 1, pain/discomfort on Factor 2, and the remaining items on Factor 3. EQ-HWB items loaded on Factors 1, 2, 3, and 5. All ICECAP-A items loaded exclusively on Factor 4. Most factor loadings were strong (>0.55), with a few exceptions in the EQ-5D-5L and ICECAP-A showing weak loadings (0.33-0.54). The EQ-HWB 'seeing' item did not sufficiently load on any of the factors and exhibited high uniqueness (0.8752). Some items demonstrated cross-loadings, for example, the EQ-HWB item 'unsafe' loading on both Factors 1 and 3, and the ICECAP-A 'stability' item loading on Factors 1 and 4.

DISCUSSION

This study contributes to the growing evidence base for the EQ-HWB's psychometric performance across distinct settings and populations[36-45], with its novel contribution lying in the first direct comparison to ICECAP-A—providing critical insights into cross-instrument validity and contextual applicability of these measures in Ireland.

One of the key findings was the absence of ceiling effects at the instrument level for both the EQ-HWB and EQ-HWB-S, in contrast to the ceiling effects observed in the EQ-5D-5L and ICECAP-A. This can be attributed to the EQ-HWB's broader content coverage of both health and wellbeing domains, and its higher number of items, even in the shortened EQ-HWB-S version.

The interpretation of ceiling effects, particularly at the item level, warrants careful consideration in the context of our sample characteristics. While we observed high proportions of responses in the best category for several physical function and sensory items (such as mobility, self-care, and hearing), this may reflect our relatively young, generally healthy sample rather than limitations in the measurement properties. For instance, the ceiling effects observed in mobility and hearing items are expected in a younger population with typically good physical functioning and intact sensory capabilities. This highlights the importance of distinguishing between ceiling effects that indicate measurement limitations versus those that appropriately reflect the health status of the population being studied.

Our analysis also showed strong correlations between conceptually similar domains of the EQ-5D-5L and EQ-HWB, supporting their convergent validity. However, the correlations between the EQ-HWB and ICECAP-A items were generally weaker than anticipated. For example, the EQ-HWB 'control' item showed only a moderate correlation with ICECAP-A 'autonomy', despite both concepts relating to independence. This finding suggests that while both measures aim to assess aspects of wellbeing, they may be capturing distinct dimensions or conceptualizing similar constructs differently.

All measures demonstrated good known-groups validity for self-reported health status, EQ VAS scores, and the presence of 'ongoing medical conditions'. However, the EQ-5D-5L showed the largest effect sizes in discriminating between these groups, followed closely by the EQ-HWB and EQ-HWB-S, with the ICECAP-A showing slightly smaller, but still significant, effect sizes. This suggests that while all measures are sensitive to health-related factors, the EQ-5D-5L and EQ-HWB may be particularly valuable in clinical settings or health-focused assessments.

Comparing our results with existing psychometric testing of the EQ-HWB and ICECAP-A alone,[29, 46-51] we found largely consistent patterns in terms of validity. However, this study provides novel insights into the relationship between these measures, particularly in terms of their factor structure and the distinct aspects of wellbeing they capture. These findings have important implications for the selection and use of health and wellbeing measures in various contexts.

The choice between these measures should be guided by the specific goals of the assessment and the population being studied. The EQ-5D-5L remains a strong choice for health-focused evaluations, particularly when discrimination between health states is a priority. The EQ-HWB offer a broader coverage of both health and wellbeing aspects, with lower ceiling, making them suitable for more comprehensive assessments of overall quality of life. The ICECAP-A, while showing some overlap with health-related measures, appears to capture unique aspects of capability wellbeing that are not fully addressed by the other measures. The ICECAP-A and EQ-HWB can be considered supplementary measures, each capturing distinct aspects of (health and) wellbeing.

The recently proposed modifications to EQ-HWB in terms of item and response level wording, ordering of items and particularly the shift from positive to negative framing for three items, could potentially strengthen correlations with corresponding ICECAP-A items and improve its overall performance in measuring wellbeing. These changes may reduce observed item level floor effects, change factor structure, and enhance sensitivity. The maintenance of the same frequency scale for positively framed items, without explicit notification of the frame shift, could potentially introduce acquiescence bias.[52] This potential bias might partially explain the observed floor effects in these items. However, empirical testing is necessary to confirm these hypotheses and reassess psychometric properties once the descriptive system is finalized.

While this study provides valuable insights, several limitations should be noted. First, the overrepresentation of younger adults and females in our sample may have influenced the findings in several ways. First, younger respondents typically have different health experiences and priorities compared to older adults, likely affecting how the items were interpreted and scored. The unexpected reverse age gradient in health and wellbeing scores (i.e. lower health and wellbeing in younger adults) might reflect this, as our younger cohort may not be representative of the general young adult population. Second, the gender imbalance towards females could have impacted our results since research shows gender differences in health reporting patterns, symptom awareness, and healthcare-seeking behaviour. This may have particularly affected items related to mental health, social relationships, and overall wellbeing assessment. Having said that, prior research showed that residents of Ireland attach a higher decrement to pain/discomfort and anxiety/depression than do other populations.[20] Additionally, the underrepresentation of older adults means we may have captured a limited range of chronic conditions and functional limitations that typically increase with age, potentially affecting discriminative ability across different health states. Second, the reliance on self-reported medical conditions without clinical verification or severity assessment restricted our ability to evaluate performance across different levels of health impairment. Third, the relatively small item pool may have influenced the factor analysis outcomes, as evidenced by the EQ-HWB sight item not loading on any factor. Fourth, the fixed order of administration could have introduced ordering effects.

This study provides evidence for multiple aspects of validity, yet further research is needed to assess test-retest reliability, responsiveness to change over time and performance in specific demographic and clinical populations, including social care settings or among older adults.

CONCLUSION

This study provides valuable insights into the comparative psychometric properties of the EQ-HWB long and short forms, EQ-5D-5L, and ICECAP-A in an Irish context. While all measures demonstrated generally good psychometric properties, they each offer unique strengths in measuring different aspects of health and wellbeing. The EQ-5D-5L excels in health state discrimination but is constrained by ceiling effects in less severe health states, the EQ-HWB and EQ-HWB-S successfully address ceiling effects through broader coverage but this comes at the cost of increased respondent burden, and the ICECAP-A captures distinct aspects of capability wellbeing but shows limited sensitivity in discriminating health states. These findings suggest that while these instruments measure different constructs, methodological challenges remain in their application, including content overlap, response burden, and sensitivity to change. Further validation studies are needed to establish clear guidance on when and how to use these instruments in economic evaluations, particularly regarding the value added by the EQ-HWB's broader but potentially more burdensome approach.

REFERENCES

1. Peasgood T, Mukuria C, Carlton J, et al. What is the best approach to adopt for identifying the domains for a new measure of health, social care and carer-related quality of life to measure quality-adjusted life years? Application to the development of the EQ-HWB? The European Journal of Health Economics. 2021; 22: 1067-81.

2. Brazier J, Peasgood T, Mukuria C, et al. The EQ-HWB: Overview of the Development of a Measure of Health and Wellbeing and Key Results. Value Health. 2022; 25: 482-91.

3. Mukuria C, Peasgood T, McDool E, et al. Valuing the EQ Health and Wellbeing Short Using Time Trade-Off and a Discrete Choice Experiment: A Feasibility Study. Value in Health. 2023; 26: 1073-84.

4. Busschbach JJ. Richtlijn voor het uitvoeren van economische evaluaties in de gezondheidszorg (versie 2024), verdiepingsmodule QALY en kwaliteit van Leven metingen. 2024.

5. Sutton EJ, Coast J. Development of a supportive care measure for economic evaluation of end-of-life care using qualitative methods. Palliat Med. 2014; 28: 151-7.

6. Grewal I, Lewis J, Flynn T, et al. Developing attributes for a generic quality of life measure for older people: preferences or capabilities? Soc Sci Med. 2006; 62: 1891-901.

7. Al-Janabi H, Flynn TN, Coast J. Development of a self-report measure of capability wellbeing for adults: the ICECAP-A. Qual Life Res. 2012; 21: 167-76.

8. Coast J. A capability based focus for economic evaluation. https://www.youtube.com/watch?v=elF3GhwZzgY&feature=youtu.be, 2016.

9. Afentou N, Kinghorn P. A Systematic Review of the Feasibility and Psychometric Properties of the ICEpop CAPability Measure for Adults and Its Use So Far in Economic Evaluation. Value in Health. 2020; 23: 515-26.

10. Proud L, McLoughlin C, Kinghorn P. ICECAP-O, the current state of play: a systematic review of studies reporting the psychometric properties and use of the instrument over the decade since its publication. Qual Life Res. 2019; 28: 1429-39.

11. National Institute for Health and Care Excellence (NICE). NICE health technology evaluations: the manual. London: NICE, 2022.

12. Goranitis I, Coast J, Day E, et al. Maximizing Health or Sufficient Capability in Economic Evaluation? A Methodological Experiment of Treatment for Drug Addiction. Medical decision making : an international journal of the Society for Medical Decision Making. 2017; 37: 498-511.

13. Himmler S, van Exel J, Brouwer W. Estimating the monetary value of health and capability well-being applying the well-being valuation approach. Eur J Health Econ. 2020; 21: 1235-44.

14. Bailey C, Dalziel K, Constable L, et al. The performance of the EQ-HWB-S as a measure of quality-of-life of caregivers in families that have experienced adverse events. The European Journal of Health Economics. 2024.

15. Kuharic M, Mulhern B, Sharp LK, et al. Comparison of EQ Health and Well-Being Long and Short With Other Preference-Based Measures Among United States Informal Caregivers. Value in Health. 2024.

16. Engel L, Kosowicz L, Bogatyreva E, et al. Face Validity of Four Preference-Weighted Quality-of-Life Measures in Residential Aged Care: A Think-Aloud Study. Patient. 2023.

17. Zigler CK, Coles T. Clinical Outcome Assessments Are Never "Validated". Value in Health. 2024; 27: 1494-96.

18. Kinchin I, Bliemer, M., Rose, J., de Bekker - Grob, E., Dams, J., Coast, J. Deriving utility values for the ICECAP-SCM in Ireland: Influence of attribute design and text simplification on preference measurements. Trinity College Dublin, Working paper.

19. Herdman M, Gudex C, Lloyd A, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). Qual Life Res. 2011; 20: 1727-36.

20. Hobbins A, Barry L, Kelleher D, et al. Utility Values for Health States in Ireland: A Value Set for the EQ-5D-5L. Pharmacoeconomics. 2018; 36: 1345-53.

21. Devlin NJ, Shah KK, Feng Y, et al. Valuing health-related quality of life: An EQ-5D-5L value set for England. Health Economics. 2018; 27: 7-22.

22. van Hout B, Janssen MF, Feng Y-S, et al. Interim Scoring for the EQ-5D-5L: Mapping the EQ-5D-5L to EQ-5D-3L Value Sets. Value in Health. 2012; 15: 708-15.

23. Flynn TN, Huynh E, Peters TJ, et al. Scoring the Icecap-a Capability Instrument. Estimation of a UK General Population Tariff. Health Economics. 2015; 24: 258-69.

24. Peasgood T, Mukuria C, Brazier J, et al. Developing a New Generic Health and Wellbeing Measure: Psychometric Survey Results for the EQ-HWB. Value Health. 2022; 25: 525-33.

25. Terwee CB, Bot SDM, de Boer MR, et al. Quality criteria were proposed for measurement properties of health status questionnaires. Journal of Clinical Epidemiology. 2007; 60: 34-42.

26. Cheng LJ, Pan T, Chen LA, et al. The Ceiling Effects of EQ-5D-3L and 5L in General Population Health Surveys: A Systematic Review and Meta-Analysis. Value in Health. 2024; 27: 986-97.

27. Cohen J. A power primer. Psychological Bulletin. 1992; 112: 155-59.

28. Cherepanov D, Palta M, Fryback DG, et al. Gender differences in health-related quality-of-life are partly explained by sociodemographic and socioeconomic variation between adult men and women in the US: evidence from four US nationally representative data sets. Quality of Life Research. 2010; 19: 1115-24.

29. Mitchell PM, Al-Janabi H, Byford S, et al. Assessing the validity of the ICECAP-A capability measure for adults with depression. BMC Psychiatry. 2017; 17: 46.

30. Ahn S-K, Seo H-J, Choi M-J. Trends and regional distribution in health-related quality of life across sex and employment status: a repeated population-based cross-sectional study. Journal of Occupational Health. 2024; 66.

31. Hobbins A, Barry L, Kelleher D, et al. The health of the residents of Ireland: Population norms for Ireland based on the EQ-5D-5L descriptive system ? a cross sectional study [version 1; peer review: 1 approved, 3 approved with reservations]. HRB Open Research. 2018; 1.

32. Hansen T, Blekesaune M. The age and well-being “paradox”: a longitudinal and multidimensional reconsideration. European Journal of Ageing. 2022; 19: 1277-86.

33. Fritz CO, Morris, P. E., & Richler, J. J. Effect size estimates: Current use, calculations, and interpretation. Journal of Experimental Psychology: General. 2012; 141: 2-18.

34. Rencz F, Janssen MF. Testing the Psychometric Properties of 9 Bolt-Ons for the EQ-5D-5L in a General Population Sample. Value in Health. 2024; 27: 943-54.

35. Comrey AL, Lee, H.B. A first course in factor analysis. 2nd ed. Hillsdale, NJ, US: Lawrence Erlbaum Associates, 1992.

36. McDool E, Mukuria C, Peasgood T. Psychometric Performance of the EQ Health and Wellbeing Short in a United Kingdom Population Sample. Value in Health. 2024; 27: 1215-24.

37. Lee P, Engel L, Lubetkin E, et al. Exploring the Comparability Between EQ-5D and the EQ Health and Wellbeing in the General Australian Population. Value in Health. 2024; 27: 508-17.

38. Kuharić M, Pickard AS, Mukuria C, et al. The Measurement Properties of the EQ-HWB and the EQ-HWB-S in Italian Population: A Comparative Study With EQ-5D-5L. Value in Health. 2024; 27: 955-66.

39. Long C, Mao Z, Yang Z. A Head-to-Head Comparison of EQ-HWB and EQ-5D-5L in Patients, Carers, and General Public in China. Value in Health. 2024; 27: 848-56.

40. Bailey C, Dalziel K, Constable L, et al. The performance of the EQ-HWB-S as a measure of quality-of-life of caregivers in families that have experienced adverse events. The European Journal of Health Economics. 2025; 26: 7-21.

41. Kuharic M, Mulhern B, Sharp LK, et al. Comparison of the EQ-HWB and EQ-HWB-S With Other Preference-Based Measures Among United States Informal Caregivers. Value in Health. 2024; 27: 967-77.

42. Pangestu S, Purba FD, Setyowibowo H, et al. The Psychometric Properties of the EQ-HWB and EQ-HWB-S in Patients With Breast Cancer: A Comparative Analysis With EQ-5D-5L, FACT-8D, and SWEMWBS. Value in Health. 2025; 28: 449-59.

43. Pangestu S, Purba FD, Setyowibowo H, et al. Associations between financial toxicity, health-related quality of life, and well-being in Indonesian patients with breast cancer. Quality of Life Research. 2025.

44. Zhang G, Yang Z, Luo N, et al. Comparing the Measurement Properties of the Preliminary Version of the EuroQol Health and Well-Being and EQ-5D-5L in Patients, Healthy General Public, and Caregivers. Value in Health Regional Issues. 2025; 47.

45. Zhou W, Ding B, Busschbach J, et al. EQ-5D-5L or EQ-HWB-S: Which is the Better Instrument for Capturing Spillover Effects in Parental Carers of Children with COVID-19? PharmacoEconomics. 2025.

46. Kuharic M. EQ-HWB Goes to Italy: Comparing Measurement Properties of the EQ Health and Wellbeing Experimental Version (Long and Short) and the EQ-5D-5L in the Italian Population. EuroQol ECR meeting. Milan, Italy, 2023.

47. Kuharic M, Monteiro, A.L., A. Pickard, S. EQ-HWB-S and EQ-HWB as an outcome measure in mental health and well-being: a comparison to the EQ-5D-5L, EQ-5D-3L and the Short Warwick-Edinburgh Mental Well-being Scale 39th EuroQol Plenary. Chicago 2022.

48. Keeley T, Coast J, Nicholls E, et al. An analysis of the complementarity of ICECAP-A and EQ-5D-3 L in an adult population of patients with knee pain. Health and Quality of Life Outcomes. 2016; 14: 36.

49. Masutti S, Falivena C, Purba FD, et al. Content validity of the EQ-HWB and EQ-HWB-S in a sample of Italian patients, informal caregivers and members of the general public. Journal of Patient-Reported Outcomes. 2024; 8: 36.

50. McDool E, Mukuria C, Peasgood T. Psychometric Performance of the EQ Health and Wellbeing Short in a United Kingdom Population Sample. Value in Health.

51. Rencz F, Mitev AZ, Jenei B, et al. Measurement properties of the ICECAP-A capability well-being instrument among dermatological patients. Quality of Life Research. 2022; 31: 903-15.

52. Choi BC, Pak AW. A catalog of biases in questionnaires. Prev Chronic Dis. 2005; 2: A13.

53. Central Statistics Office. Census of Population 2016 - Profile 3 An Age Profile of Ireland. 2016.

Table 1: Participant characteristics (n = 1220)

Demographic characteristics	Frequency (%)	National statistics
Age		
18-24 years	302 (24.75)	8.2%
25-34 years	325 (26.64)	13.8%
35-44 years	278 (22.79)	15.7%
45-54 years	198 (16.23)	13.1%
55-64 years	80 (6.56)	10.7%
65 or older	36 (2.95)	13.3%
Prefer not to answer	1 (0.08)	
Gender		
Female	815 (66.80)	50.6%
Male	395 (32.38)	49.2%
Non-binary	6 (0.49)	
Prefer not to answer	4 (0.33)	
Partnership status		
Married / Civil Partnered / Living together	635 (52.05)	
Single	511 (41.89)	
Divorced / Separated	47 (3.85)	
Widowed	11 (0.90)	
Other	3 (0.25)	
Prefer not to answer	13 (1.07)	
Employment status		
Employed or self-employed	748 (61.31)	
Retired	40 (3.28)	
Student	132 (10.82)	
Looking after home or family	193 (15.82)	
Long-term sick or disabled	59 (4.84)	
Prefer not to answer	48 (3.93)	
Highest level of education		
Some post-primary education or less	69 (5.66)	
Leaving certificate	292 (23.93)	

Post-Leaving certificate vocational training	148 (12.13)
Completed some third-level education	177 (14.51)
Higher certificate	129 (10.57)
Bachelor/ MSc/ PhD	391 (32.05)
None of the above	1 (0.08)
Prefer not to answer	13 (1.07)
Ethnicity	
White Irish	951 (77.95)
White Irish traveller	25 (2.05)
Any other white background	129 (10.57)
Black or Black Irish - African	30 (2.46)
Black or Black Irish - any other Black	12 (0.98)
Asian or Asian Irish - Chinese	7 (0.57)
Asian or Asian Irish - any other Asian	26 (2.13)
Other (incl. mixed background)	24 (1.97)
Prefer not to answer	16 (1.31)
Country of birth	
Ireland	942 (77.21)
Other	278 (22.79)
Region	
Dublin	441 (36.15)
South East	152 (12.46)
South West	140 (11.48)
Mid East	134 (10.98)
Border	108 (8.85)
Midlands	98 (8.03)
West	76 (6.23)
Mid West	71 (5.82)
Are you currently providing help or support?	
Yes	450 (36.89)
Not currently, but I have in the past	261 (21.39)
No	496 (40.66)
Prefer not to answer	13 (1.07)

In total, how many hours did you spend on helping or supporting this person (out of *n*=450)	
1 to 19	218 (48.44)
20 to 49	127 (28.22)
50 or more	87 (19.33)
Don't know	18 (4.00)
Do you have dependent children of any age?	
Yes	648 (53.11)
No	555 (45.49)
Prefer not to answer	17 (1.39)
General health status	
Excellent	166 (13.61)
Very good	472 (38.69)
Good	389 (31.89)
Fair	162 (13.28)
Poor	31 (2.54)
EQ VAS	
EQ VAS =< 80	749 (61.39)
EQ VAS > 80	471 (38.61)
Ongoing medical condition	
Anxiety	464 (38.03)
Arthritis	85 (6.97)
Asthma	152 (12.46)
Back problem	211 (17.30)
Cancer	16 (1.31)
Chronic obstructive pulmonary disease (COPD)	21 (1.72)
Chronic pain	69 (5.66)
Depression	227 (18.61)
Diabetes	46 (3.77)
Heart, stroke, and vascular disease	13 (1.07)
Hypertension	95 (7.79)
High cholesterol	103 (8.44)
Osteoporosis	26 (2.13)

Kidney disease	11 (0.90)
Other mental health and behavioural condition	95 (7.79)
Other	68 (5.57)
No ongoing medical conditions	383 (31.39)
Any	837 (68.61)

Note: *Central Statistics Office. Census of Population 2016 - Profile 3 An Age Profile of Ireland. 2016; https://www.cso.ie/en/releasesandpublications/ep/p-cp3oy/cp3/assr/. Accessed 15 June, 2023.[53]

Table 2: Descriptive statistics, floor, and ceiling effect

Instrument/index	Number of items	Minimum (floor) n (%)	Maximum (ceiling) n (%)	Mean (SD)	Min-Max (observed)	Min-Max (theoretical)	Median	Mode
EQ-5D-5L	5	0	275 (22.54)					
EQ-5D-5L Index (England; Devlin et al)				0.830 (0.179)	-0.175 to 1	-0.285 to 1	0.872	1
EQ-5D-5L Index (UK crosswalk; van Hout et al)				0.760 (0.224)	-0.414 to 1	-0.594 to 1	0.791	1
EQ-5D-5L Index (Ireland; Hobbins et al)				0.770 (0.269)	-0.754 to 1	-0.974 to 1	0.852	1
EQ VAS	-	2 (0.16)	44 (3.61)	71.60 (19.63)	0 to 100	0 to 100	77	80
EQ-HWB	25	0	32 (2.62)					
EQ-HWB-LSS transformed				26.9 (16.04)	0 to 78	0 to 100	25	12
EQ-HWB-S	9	0	81 (6.64)					
EQ-HWB-S Index (UK pilot; Mukuria et al)				0.755 (0.211)	-0.204 to 1	-0.384 to 1	0.819	1
ICECAP-A	5	0	178 (14.59)					
ICECAP-A Index (UK)				0.807 (0.178)	0.077 to 1	0 to 1	0.861	1

Note: Higher scores represent better health or wellbeing except for EQ-HWB LSS

Table 3: Distribution of item response for EQ-5D-5L, EQ-HWB, EQ-HWB-S, and ICECAP-A

Instrument	Item	Ceiling, %				Floor┼, %
		1	2	3	4	5
EQ-5D-5L[a]	Mobility	**78.85**	13.61	5.49	1.48	0.57
	Self-care	**85.98**	7.54	4.43	1.64	0.41
	Usual activities	**70.41**	18.28	8.28	2.46	0.57
	Pain/discomfort	44.92	37.30	12.46	4.02	1.31
	Anxiety/depression	38.20	34.10	17.54	7.62	2.54
EQ-HWB[b]	Seeing (difficulty)	57.21	25.41	12.13	4.84	0.41
	Hearing (difficulty)	**73.52**	14.92	7.87	3.28	0.41
	Getting around inside and outside (difficulty)*	**79.18**	9.59	7.79	2.54	0.90
	Daily activities (difficulty)*	66.15	20.66	8.52	4.10	0.57
	Washing, using the toilet, getting dressed, eating, or caring for your appearance (difficulty)	**78.93**	9.75	7.05	3.52	0.74
	Problems w/sleep (frequency)	26.23	34.02	22.21	11.89	5.66
	Feeling exhausted (frequency)*	16.64	29.59	27.62	18.85	7.30
	Feeling lonely (frequency)*	35.98	25.98	21.56	11.64	4.84
	Feeling that people did not support (frequency)	36.89	27.62	22.54	9.34	3.61
	Trouble remembering (frequency)	46.97	27.54	15.82	7.30	2.38
	Trouble concentrating/thinking clearly (frequency)*	35.08	31.15	21.80	9.18	2.79
	Feeling anxious (frequency)*	27.54	27.54	24.59	14.67	5.66
	Feeling unsafe (frequency)	63.28	18.69	11.89	3.69	2.46
	Feeling frustrated (frequency)	24.02	32.38	26.48	13.36	3.77

	Feeling sad/depressed (frequency)*	30.41	30.41	22.21	12.30	4.67
	Feeling nothing to look forward to (frequency)	40.41	25.41	18.69	9.84	5.66
	Feeling no control over day-to-day life (frequency) *	44.51	24.34	18.03	9.59	3.52
	Feeling unable to cope with day-to-day life (frequency)	48.20	24.18	15.74	7.87	4.02
	Feeling accepted (frequency)[d]	25.16	21.80	21.80	17.13	14.10
	Feeling good (frequency)[d]	15.08	22.62	29.18	20.49	12.62
	Could do things wanted (frequency)[d]	20.82	22.46	27.62	18.52	10.57
	Physical pain (frequency)	35.41	32.87	19.02	9.26	3.44
	Physical pain (severity)*	38.20	37.87	17.05	5.57	1.31
	Physical discomfort (frequency)	47.70	29.18	16.72	5.16	1.23
	Physical discomfort (severity)	41.39	39.34	14.26	4.02	0.98
ICECAP-A[c,d]	Stability (feeling settled and secure)	30.98	42.95	20.25	5.82	na
	Attachment (love, friendship and support)	44.34	34.59	19.10	1.97	na
	Autonomy (being independent)	51.31	35.25	11.72	1.72	na
	Achievement (achievement and progress)	31.48	44.59	20.16	3.77	na
	Enjoyment (enjoyment and pleasure)	38.03	38.69	20.82	2.46	na

Note: EQ-HWB = EQ Health and Wellbeing; w/ = with; ICECAP-A - ICEpop CAPability measure for Adults; na = not applicable

[a] EQ-5D-5L response scale: 1 = no problems, 2 = slight problems, 3 = moderate problems, 4 = severe problems, 5 = unable/extreme problems.

[b] EQ-HWB response scale: difficulty scale (1 = no difficulty, 2 = slight difficulty, 3 = some difficulty, 4 = a lot of difficulty, 5, unable); frequency scale (1 = none of the time, 2 = only occasionally, 3 = sometimes, 4 = often, 5 = most or all of the time); severity scale (1 = no pain, 2 = mild pain, 3 = moderate pain, 4 = severe pain, 5 = very severe pain)

[c] ICECAP-A response scale: 1 = full capability (can have, able) to 4 = no capability (cannot have/achieve, unable)

[d] Rescaled to ensure consistency across the response characteristics

* EQ-HWB-S items

┼ Level 4 was considered the floor for the ICECAP-A

Bolded ceiling effects >70% responses in the top category

Table 4: Correlation coefficients (Pearson) between EQ-5D-5L, EQ-HWB, ICECAP-A indices (*n*=1220)

Instrument	EQ-5D-5L Index (van Hout et al)	EQ-5D-5L Index (Hobbins et al)	EQ VAS	EQ-HWB LSS	EQ-HWB-S Index (Mukuria et al)	ICECAP-A Index
EQ-5D-5L Index (Devlin et al)	**0.970**	**0.977**	0.474	**-0.719**	**0.759**	**0.525**
EQ-5D-5L Index (van Hout et al)	-	**0.957**	0.467	**-0.705**	**0.740**	**0.514**
EQ-5D-5L Index (Hobbins et al)		-	0.466	**-0.710**	**0.746**	**0.520**
EQ VAS			-	**-0.505**	0.489	0.461
EQ-HWB LSS				-	**-0.922**	**-0.648**
EQ-HWB-S Index (Mukuria et al)					-	**0.604**

Note: Higher scores represent better HRQoL/wellbeing for all measures except for EQ-HWB LSS. All Pearson's correlation coefficient were statistically significant (P <0.001). Correlations were interpreted according to Cohen's guidelines, i.e., "strong" (≥0.50), "moderate" (0.30-0.49), "weak" (0.10-0.29), and "none" (0-0.09). (+) indicates positive direction; (-) negative direction. Strong correlations are **bolded**.

Table 5: Known-group validity of the EQ-5D-5L, EQ-HWB, EQ-HWB-S and ICECAP-A

Characteristic	Sample (N=1220)	Instrument EQ-5D-5L Index (Ireland)	EQ-5D-5L Index (UK crosswalk)	EQ-5D-5L Index (England)	EQ-HWB-LSS[a]	EQ-HWB-S Index (UK)	ICECAP-A Index (UK)
	n	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
Employment[1]							
Employed/ Self-employed	748	0.810 (0.238)	0.793 (0.198)	0.857 (0.157)	24.93 (15.220)	0.781 (0.199)	0.828 (0.164)
Retired	40	0.751 (0.368)	0.739 (0.310)	0.807 (0.253)	22.50 (19.060)	0.789 (0.250)	0.798 (0.188)
Student	132	0.734 (0.258)	0.739 (0.211)	0.816 (0.165)	29.89 (15.410)	0.721 (0.201)	0.807 (0.166)
Looking after home, family	193	0.740 (0.267)	0.733 (0.218)	0.810 (0.175)	29.19 (16.420)	0.728 (0.205)	0.782 (0.190)
Long-term sick or disabled	59	0.438 (0.355)	0.459 (0.271)	0.581 (0.238)	40.05 (16.790)	0.546 (0.265)	0.639 (0.196)
		KW 92.42* **ES 0.08**	KW 97.02* **ES 0.08**	KW 97.44* **ES 0.08**	KW 57.00* ES 0.05	KW 69.16* **ES 0.06**	KW 56.07* ES 0.05
Education[2]							
Bachelor/ MSc/ PhD	391	0.808 (0.255)	0.798 (0.206)	0.857 (0.169)	24.02 (15.580)	0.788 (0.199)	0.823 (0.170)
Third-level education	177	0.766 (0.275)	0.756 (0.227)	0.829 (0.185)	26.88 (15.720)	0.750 (0.213)	0.810 (0.168)
Higher certificate	129	0.778 (0.242)	0.764 (0.205)	0.835 (0.157)	27.86 (15.990)	0.746 (0.204)	0.806 (0.188)
Leaving certificate	292	0.745 (0.274)	0.739 (0.228)	0.815 (0.180)	28.34 (16.250)	0.740 (0.219)	0.800 (0.184)
Post-leaving cert training	148	0.729 (0.299)	0.717 (0.250)	0.798 (0.200)	30.30 (16.510)	0.713 (0.226)	0.770 (0.179)
Post-primary education	69	0.749 (0.262)	0.731 (0.234)	0.808 (0.181)	27.61 (15.270)	0.749 (0.203)	0.814 (0.176)
		KW 21.25* ES 0.02	KW 25.30* ES 0.02	KW 23.63* ES 0.02	KW 24.36* ES 0.02	KW19.80* ES 0.01	KW 11.22‡ ES 0.01
Caregiving role[3]							

Yes	450	0.728 (0.289)	0.720 (0.234)	0.798 (0.192)	30.52 (16.260)	0.709 (0.224)	0.793 (0.185)
Not currently, in the past	261	0.758 (0.264)	0.751 (0.213)	0.825 (0.167)	27.14 (16.350)	0.755 (0.207)	0.800 (0.171)
No	496	0.819 (0.232)	0.805 (0.199)	0.865 (0.159)	23.33 (14.850)	0.800 (0.189)	0.825 (0.174)
	1207	KW 37.65* ES 0.03	KW 43.73* ES 0.04	KW 42.35* ES 0.03	KW 48.46* ES 0.04	KW 45.95* ES 0.04	KW 11.00† ES 0.01
General health status							
Excellent	166	0.858 (0.229)	0.839 (0.205)	0.893 (0.154)	21.01 (14.080)	0.824 (0.182)	0.877 (0.139)
Very good	472	0.817 (0.230)	0.803 (0.201)	0.863 (0.156)	24.15 (14.990)	0.794 (0.180)	0.840 (0.163)
Good	389	0.778 (0.235)	0.759 (0.198)	0.836 (0.148)	27.49 (15.390)	0.753 (0.199)	0.802 (0.165)
Fair/ Poor	193	0.561 (0.344)	0.586 (0.250)	0.682 (0.227)	37.50 (16.470)	0.602 (0.254)	0.674 (0.198)
		KW 165.60* **ES 0.13**	KW 174.85* **ES 0.14**	KW 172.73* **ES 0.14**	KW 110.51* **ES 0.09**	KW 122.45* **ES 0.10**	KW 143.93* **ES 0.12**
EQ VAS							
=< 80	749	0.700 (0.294)	0.699 (0.233)	0.782 (0.193)	31.75 (15.870)	0.697 (0.222)	0.758 (0.187)
> 80	471	0.881 (0.175)	0.856 (0.169)	0.906 (0.122)	19.18 (13.020)	0.847 (0.153)	0.884 (0.128)
		Z -13.91* **ES 0.47**	Z -13.62* **ES 0.46**	Z -13.95* **ES 0.47**	Z -13.59* **ES 0.46**	Z -13.15* **ES 0.45**	Z -12.63* **ES 0.43**
Ongoing medical condition							
None	383	0.914 (0.144)	0.884 (0.159)	0.926 (0.114)	18.68 (13.070)	0.859 (0.151)	0.878 (0.135)
Any	837	0.704 (0.287)	0.702 (0.226)	0.786 (0.187)	30.66 (15.880)	0.707 (0.218)	0.774 (0.185)
		Z 16.69* **ES 0.59**	Z 15.39* **ES 0.54**	Z 16.06* **ES 0.57**	Z 12.51* **ES 0.45**	Z 13.22* **ES 0.47**	Z 10.00* **ES 0.36**

Note: [a] Transformed on a scale 0-100. Higher scores represent better HRQoL/wellbeing for all measures except for EQ-HWB LSS.

[1] Excluded Prefer not to answer (*n*=48)

[2] Excluded Prefer not to answer (*n*=13) and none of the above (*n*=1)

[3] Excluded Prefer not to answer (*n*=13)

*$p < 0.001$

†$p < 0.01$

‡$p < 0.05$

Kruskal–Wallis H effect size (ES): "small" 0.01-0.059, "moderate" 0.06-0.139, "large" ≥0.14; Mann–Whitney U effect size (ES): "small" 0.11-0.30, "moderate" 0.31-0.50, "large" ≥0.5. **Bolded** results indicate a <u>large-moderate</u> effect size.

Table 6: Results of exploratory factor analysis (5-factor-model)

Instrument	Item	Factor 1 - psychosocial health	Factor 2 - pain and discomfort	Factor 3 - sensory and physical functioning	Factor 4 - capability wellbeing	Factor 5 - positive psychological states	U
EQ-5D-5L[a]	Mobility			0.4465			0.6153
	Self-care			0.6890			0.5229
	Usual activities			0.5411			0.4492
	Pain/discomfort		0.7057				0.4040
	Anxiety/depression	0.6617					0.4961
EQ-HWB[b]	Seeing (difficulty)						0.8752
	Hearing (difficulty)			0.5790			0.6840
	Getting around inside and outside (difficulty)*			0.8255			0.3745
	Daily activities (difficulty)*			0.7170			0.3157
	Washing, using the toilet, getting dressed, eating, or caring for your appearance (difficulty)			0.8299			0.3420
	Problems w/sleep (frequency)	0.4526					0.6230
	Feeling exhausted (frequency)*	0.5454					0.5237
	Feeling lonely (frequency)*	0.7040					0.3968
	Feeling that people did not support (frequency)	0.6509					0.4867
	Trouble remembering (frequency)	0.4796					0.6799
	Trouble concentrating/thinking clearly (frequency)*	0.6796					0.4786
	Feeling anxious (frequency)*	0.8397					0.3687
	Feeling unsafe (frequency)	0.5328		0.4023			0.4965
	Feeling frustrated (frequency)	0.7787					0.4154

	Feeling sad/depressed (frequency)*	0.9342				0.2408
	Feeling nothing to look forward to (frequency)	0.7931				0.3423
	Feeling no control over day-to-day life (frequency) *	0.6295				0.4608
	Feeling unable to cope with day-to-day life (frequency)	0.7176				0.3812
	Feeling accepted (frequency)[d]				0.7560	0.4127
	Feeling good (frequency)[d]				0.8318	0.2753
	Could do things wanted (frequency)[d]				0.7749	0.3867
	Physical pain (frequency)		0.8552			0.3312
	Physical pain (severity)*		0.8975			0.2649
	Physical discomfort (frequency)		0.4904			0.5801
	Physical discomfort (severity)		0.7602			0.3387
ICECAP-A[c]	Stability	0.3285		0.5149		0.4922
	Attachment			0.5609		0.6155
	Autonomy			0.4434		0.6482
	Achievement			0.7288		0.4308
	Enjoyment			0.6704		0.4042

Note: EQ-HWB = EQ Health and Wellbeing; w/ = with; ICECAP-A - ICEpop CAPability measure for Adults; na = not applicable; U = Uniqueness

[a] EQ-5D-5L response scale: 1 = no problems, 2 = slight problems, 3 = moderate problems, 4 = severe problems, 5 = unable/extreme problems.

[b] EQ-HWB response scale: difficulty scale (1 = no difficulty, 2 = slight difficulty, 3 = some difficulty, 4 = a lot of difficulty, 5, unable); frequency scale (1 = none of the time, 2 = only occasionally, 3 = sometimes, 4 = often, 5 = most or all of the time); severity scale (1 = no pain, 2 = mild pain, 3 = moderate pain, 4 = severe pain, 5 = very severe pain)

[c] ICECAP-A response scale: 1 = full capability (can have, able) to 4 = no capability (cannot have/achieve, unable)

[d] Rescaled to ensure consistency across the response characteristics

* EQ-HWB-S items

Blanks represent abs(loading)<.32